

Bellabeat Case Study: Analyzing Smart Device Usage Data

Tools: R, RStudio

Skills: Data Cleaning, Data Analysis, Data Visualization, Descriptive Statistics, Correlation Analysis

Business Task

Analyze smart device usage data to identify trends in users' activity and sleep habits. The goal is to use these insights to help Bellabeat better understand how customers use wellness technology and identify opportunities for marketing and product engagement.

Data Source

The dataset used for this analysis is the Fitbit Fitness Tracker dataset available through Kaggle. The dataset contains daily activity and sleep information collected from Fitbit users, including steps, calories burned, activity levels, and sleep measurements.

The analysis used two primary datasets:

- **Daily Activity:** 940 observations and 15 variables containing daily activity and calorie information.
- **Sleep Day:** Sleep records containing total minutes asleep and total time in bed.

The data was imported into RStudio and prepared for analysis by converting date fields and creating additional variables needed for analysis.

Tools & Process

Tools Used: R and RStudio

Data Cleaning:

- Imported Fitbit activity and sleep datasets into RStudio.
- Reviewed the structure and dimensions of the datasets.
- Converted date fields into the correct date format.

- Created a new **TimeAwake** variable by subtracting total minutes asleep from total time in bed.

Data Analysis:

- Calculated the mean and median daily steps.
- Calculated average calories burned.
- Calculated average activity minutes.
- Analyzed average steps by day of the week.
- Calculated average sleep time and time spent in bed.
- Examined the correlation between daily steps and calories burned.

Data Visualization:

- Average Daily Activity
- Average Steps by Day of Week
- Steps vs. Calories
- Average Sleep vs. Time Awake

Key Findings

- Users averaged approximately **7,638 steps per day**, with a median of approximately **7,406 steps**.
- Users burned an average of approximately **2,304 calories per day**.
- **Saturday** had the highest average number of steps at approximately **8,153 steps**.
- **Sunday** had the lowest average number of steps at approximately **6,933 steps**.
- Users averaged approximately **7 hours of sleep per night** and spent approximately **7.6 hours in bed**.
- Users spent an average of approximately **39 minutes awake while in bed**.
- The correlation between daily steps and calories burned was approximately **0.592**, indicating a moderate positive relationship. This means that users who recorded more daily steps generally also recorded higher calorie expenditure.
- Activity levels varied throughout the week, suggesting opportunities for Bellabeat to provide personalized activity reminders and wellness challenges.

Analyze

The analysis showed that users averaged approximately 7,638 steps per day and burned approximately 2,304 calories per day. Activity levels varied throughout the week, with Saturday having the highest average number of steps and Sunday having the lowest.

Sleep analysis showed that users averaged approximately 7 hours of sleep per night and spent approximately 7.6 hours in bed. Users spent approximately 39 minutes awake while in bed.

The correlation analysis showed a moderate positive relationship of 0.592 between daily steps and calories burned. This suggests that users who recorded more steps generally also recorded higher calorie expenditure, although correlation does not establish that increased steps directly caused the higher calorie expenditure.

Share

The analysis identified several patterns in users' activity and sleep behavior. Users averaged approximately 7,638 steps per day and burned an average of 2,304 calories per day. Saturday had the highest average number of steps at approximately 8,153, while Sunday had the lowest at approximately 6,933.

The analysis also showed that users averaged approximately 7 hours of sleep per night and spent approximately 7.6 hours in bed. On average, users spent about 39 minutes in bed awake.

The relationship between daily steps and calories burned showed a moderate positive correlation of 0.592. This indicates that users who recorded more daily steps generally also recorded higher calorie expenditure.

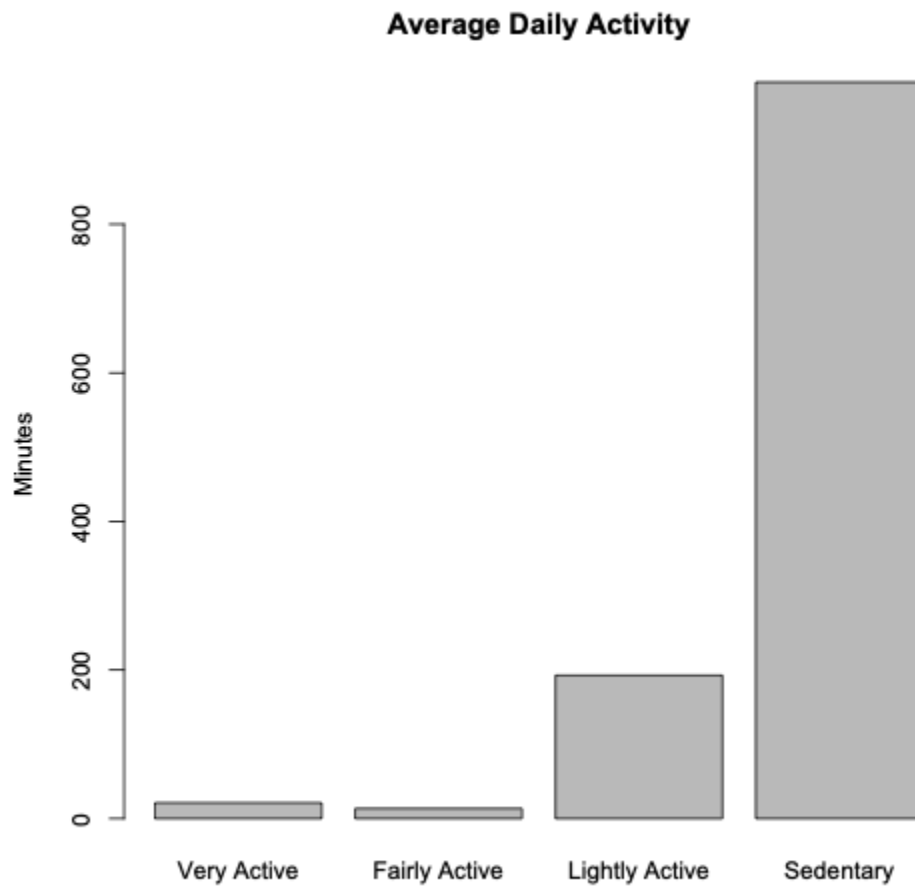
The visualizations created in R include average daily activity, average steps by day of the week, steps versus calories burned, and average sleep versus time awake.

Act

Based on the analysis, Bellabeat could consider the following opportunities:

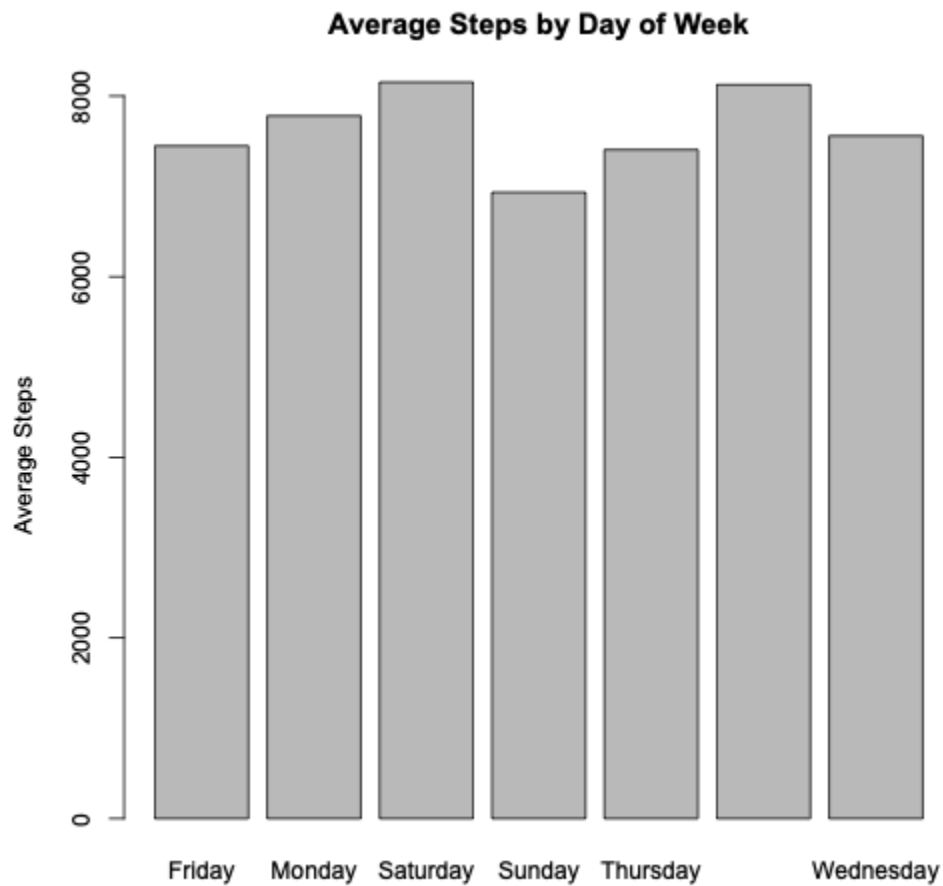
- **Encourage daily movement:** Create step challenges, activity goals, and reminders to encourage users to stay active throughout the day.
- **Focus on weekend activity:** Since Sunday had the lowest average number of steps, Bellabeat could introduce weekend-specific reminders or challenges to encourage movement.
- **Promote healthy sleep habits:** Use sleep-tracking features, personalized sleep insights, and bedtime reminders to help users monitor and improve their sleep routines.
- **Connect activity and wellness insights:** Bellabeat could combine activity, calorie, and sleep information to give users a more complete view of their wellness habits.
- **Use personalized notifications:** Provide reminders and challenges based on individual user activity and sleep patterns to encourage continued engagement with Bellabeat products.

Visualization 1: Average Daily Activity



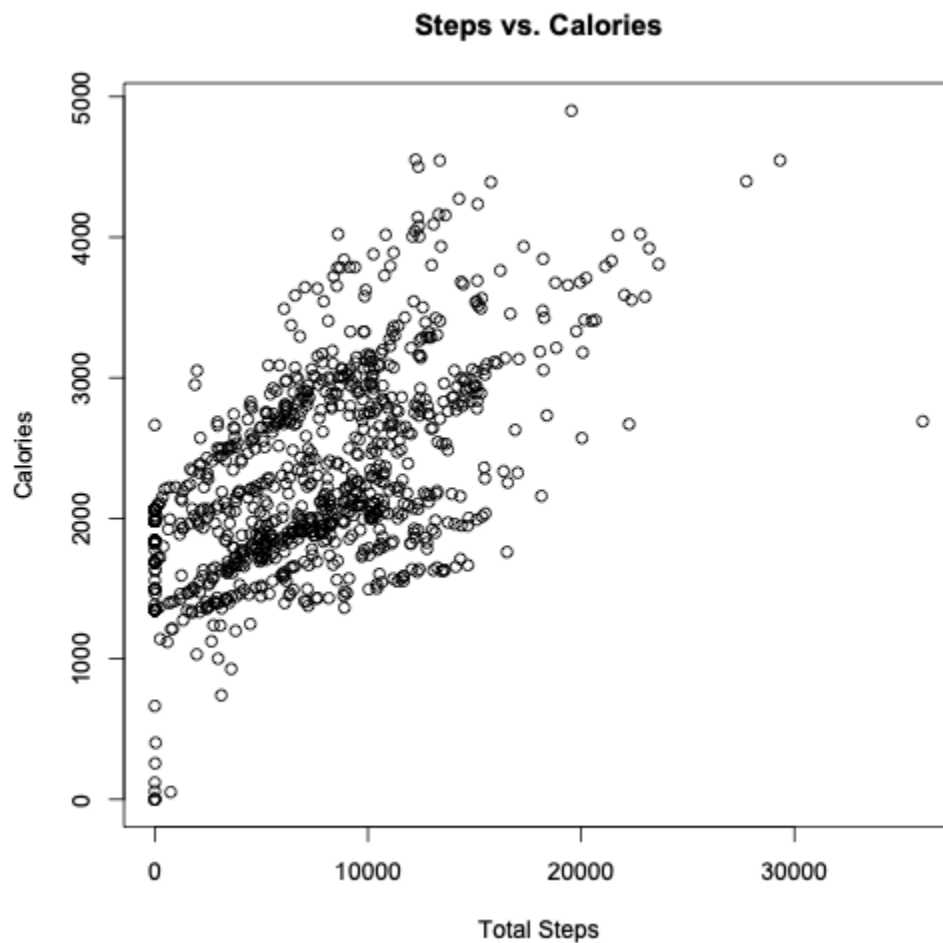
This chart shows the average amount of time users spent at different activity levels. Users spent the most time in light activity and sedentary activity compared with very active and fairly active activity.

Visualization 2: Average Steps by Day of Week



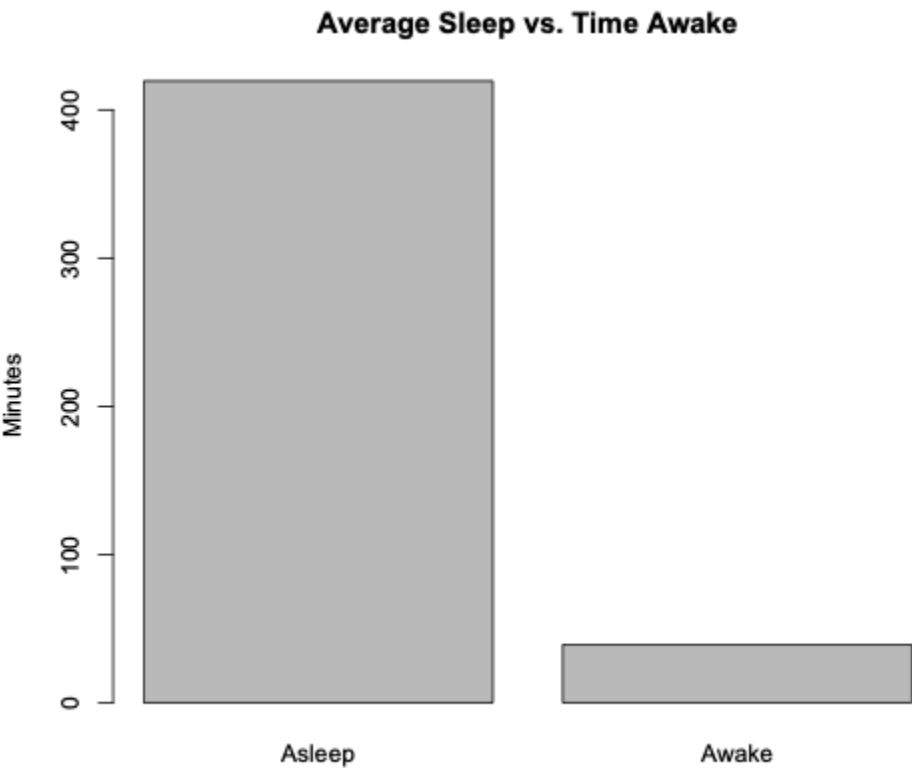
This chart compares average daily steps across the days of the week. Saturday had the highest average number of steps at approximately 8,153, while Sunday had the lowest at approximately 6,933.

Visualization 3: Steps vs. Calories



This scatter plot shows the relationship between daily steps and calories burned. The correlation was approximately 0.592, indicating a moderate positive relationship. Users who recorded more daily steps generally also recorded higher calorie expenditure.

Visualization 4: Average Sleep vs. Time Awake



This chart compares the average amount of time users spent asleep with the amount of time they spent awake while in bed. Users averaged approximately 7 hours of sleep per night and spent approximately 39 minutes awake while in bed.